MILITARY SPECIFICATION

PRINTED WIRING ASSEMBLY ILLUSTRATIONS AND MAINTENANCE DATA: REQUIREMENTS FOR AND PREPARATION OF

1. SCOPE

1.1 Scope. This specification sets forth requirements for the inclusion of printed wiring assembly maintenance data and master assembly drawing illustrations in technical publications. It specifies the type of maintenance data required and prescribes the manner in which printed wiring assemblies are to be illustrated for maintenance and test purposes.

1.2 Classification

1.2.1 Categories of technical publications. This specification shall apply to the preparation of printed wiring assembly data and master assembly drawing illustrations intended for use in technical publications in the following categories, in accordance with the requirements in this specification.

CATEGORY I Technical publications in this category are those prepared by, especially for, or at the direction of the Marine Corps. This category includes: Technical manuals, maintenance manuals, maintenance and repair handbooks and maintenance prints.

CATEGORY II Technical publications in this category are those prepared by or at the direction of agencies other than the Marine Corps, and cover equipment or systems in joint service use.

1.2.2 Types of printed wiring assemblies covered. This specification is applicable to the preparation of maintenance data for, and illustrations of, printed wiring assemblies. Where used in the specification, the term "printed wiring assembly" applies to those assemblies in which conductive, semiconductive, dielectric or magnetic patterns are formed by photoetching, silk screening, stamping, vapor deposit or similar processes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

1.2.3 Types of assemblies not covered. Printed wiring boards used within microelectronic assemblies, multilayer (cord wood) printed wiring assemblies, or assemblies in which flexible wiring is employed, in lieu of printed wiring, are not covered by this specification.

2. APPLICABLE PUBLICATIONS

2.1 The following documents of the issue in effect on the date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-21630 - Illustrations, for Stock List, and Technical Publications: Preparations of.
MIL-L-22690 - Lubrication Charts, Plates and Manuals; Preparation and Production of.
MIL-M-17993 - Marine Corps Provisioning Procedures and Technical Documentation Requirements.
MIL-T-22689 - Technical Manuals: Production of.

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings and in Technical-Type Publications.
MIL-STD-15-1 - Graphic Symbols for Electrical and Electronics Diagrams.
MIL-STD-15-3- Electrical Wiring Symbols for Architectural and Electrical Layout Drawings.

PUBLICATIONS

DEPARTMENT OF DEFENSE

DD Form 441 - Industrial Security Manual for Safeguarding Classified Information.

U. S. MARINE CORPS

SG-1 - Style Guide for the Preparation of Technical Manuals.

Source: https://assist.dla.mil -- Downloaded: 202640-23T06:45Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES OF AMERICA STANDARDS INSTITUTE

Y32.16 - 1965 - Electrical and Electronics Reference Designations.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th St., New York, N. Y. 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Technical publications requirements.

3.1.1 Requirements for category I technical publications. Technical publications in category I are prepared in accordance with the requirements set forth in the specifications, standards and publications listed in 2.1 and 2.2. In addition to such requirements, maintenance prints and those sections, chapters and volumes of technical publications in category I covering field (third and fourth) and depot (fifth) echelon maintenance shall incorporate printed wiring assembly maintenance data and master assembly drawing illustrations in accordance with 3.2, 3.3 and 3.4. When specifically required by contract, or at the written direction of the contracting officer, the requirements of this specification shall apply to the preparation of organizational (second) echelon maintenance publications. Such coverage shall be limited to that necessary to clarify second echelon maintenance test procedures, test point locations, and adjustments applicable to specific printed wiring assemblies in an equipment or system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

3.1.2 Requirements for category II technical publications. The requirements given in this specification shall apply to technical publications in category II only when specifically required by the contract, contracting agency or service, or at the written direction of the contracting officer. When this requirement is established, a category II technical publication shall be treated as a category I publication insofar as the requirements of this specification apply.

3.2 Printed wiring assembly illustrations.

3.2.1 Illustrations required. A schematic diagram and a master assembly drawing for each type of printed wiring assembly used in an equipment or system shall be included in technical publications, or portion thereof which covers field and depot maintenance. When two or more identical printed wiring assemblies are used in the same equipment or system, only one schematic and one master assembly drawing shall be required for that particular type of assembly.

3.2.2 Location and page size of illustrations. Each printed wiring assembly schematic diagram and master assembly drawing shall be prepared for inclusion as a foldout in the finished technical publication. These illustrations shall appear only on the facing pages of foldout sheets and shall not exceed the foldout image area size permitted by the specification(s) governing the preparation of illustrations for use in the particular publication in which they appear. Unless specifically prohibited by the specification(s) governing the preparation of that technical publication, printed wiring assembly schematic (foldout) diagrams and master assembly (foldout) drawings shall appear at the end of the appropriate section, chapter or volume, depending upon specific type of publication.

3.2.3 Arrangement, titling and numbering of illustrations. Each master assembly (foldout) drawing shall appear on the facing page of the foldout sheet immediately behind that on which the schematic (foldout) diagram for that particular assembly appears. A schematic diagram and associated master assembly drawing may be included on the facing page of the same foldout sheet when size permits. When this is done, the master assembly drawing is to appear to the right of the schematic diagram. Page numbers, figure numbers and figure titles for printed wiring assembly illustrations shall be in accordance with the specification(s) governing the preparation of the technical publication in which they appear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

3.3 Preparation of schematic diagrams. Terms, symbols, abbreviations, conventions and materials used in preparing printed wiring assembly schematic diagrams, shall be in compliance with those standards and specifications governing line art foldout schematic diagram illustrations intended for use in a particular technical publication. The submission of original art negatives, proofs and printed copy shall be in accordance with specifications and standards cited in the contract applicable to the technical publication in which the printed wiring assembly illustrations appear. Figure 1 illustrates a typical printed wiring assembly schematic diagram.

3.4 Master assembly drawing illustrations. Master assembly drawings appearing as illustrations in technical publications shall be in a format similar to that shown in Figure 2 and shall comply with the requirements given in 3.4.2 through 3.4.6 of the specification.

3.4.1 Definitions of terms. In addition to those given in MIL-STD-429, the following terms are used in this specification:

COMPONENT SIDE - Printed wiring assemblies having separable components attached. That side of the assembly to which all, or the majority of, separable components are attached shall be the component side, irrespective of the presence or absence of printed components in the conductor pattern(s).
- Printed wiring assemblies having no separable components attached in which printed components are incorporated. That side of the assembly on which all, or the majority of, printed components are printed.
- Printed wiring assemblies having neither separable components nor printed components. That side of the assembly to which the greatest number of external connections are made to a conductor pattern.

CONDUCTOR SIDE - That side of a printed wiring assembly opposite the component side.

3.4.2 Component side conductor pattern. The printed wiring conductor pattern (and printed components if any) on the component side of a printed wiring assembly board shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

be printed in black in printed wiring assembly master drawing illustrations. The pattern shall be a true copy of that prepared for and used in the printed wiring assembly fabrication process. The view shall be that seen from the component side of the assembly. Those portions of the conductor pattern which are partially obscured by separable components (capacitors, resistors, etc.) attached to the board shall appear as dashed line segments.

3.4.3 Component side separable components. The physical configurations of separable components attached to the component side shall be outlined by means of light weight black lines in accordance with 3.4.4. Such outlines shall appear at the approximate physical locations of the respective separable components relative to segments of the conductor pattern. All separable component (flexible wire) leads shall be shown as light weight lines extending from each of the respective components to those points on the conductor pattern at which the actual physical connections are made. Separable component schematic diagram designations shall be printed in black within or adjacent to the associated separable component outlines.

3.4.4 Relative line weights. The width of lines used to illustrate separable component outlines and separable component (flexible wire) leads shall not exceed 1/4 of that used to depict the printed wiring conductor pattern.

3.4.5 Conductor side printed wiring pattern and components. When a conductor pattern is also imprinted on the side of a printed wiring assembly board opposite that on which separable components are attached (defined as the conductor side), this pattern shall appear in a 50 ± 5 percent gray in the master assembly drawing. The conductor pattern shall appear as seen through the printed wiring assembly board from the component side and shall be properly aligned with respect to that on the component side. Separable components attached on the conductor side shall be outlined in light weight gray lines. Separable component schematic designations shall be included in accordance with 3.4.6.

3.4.6 Conductor side illustration options. The following are illustration options which may be cited in the contract or by the contracting officer. If not specified by either, the selection of these options is at the discretion of the contractor or preparing agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

(a) To emphasize separable components on the conductor side, dashed light weight gray diagonal lines may be included within the component outlines (see R28 in lower section of Figure 2).

(b) Schematic designations for components located on the conductor side of an assembly must be included. These designations, such as R4, R28, etc., may be printed in black or gray, or in gray and reversed (see "PC257" designation at lower left of Figure 2).

3.4.7 Master assembly drawing original art. When the submission of original art for figure illustrations appearing in a technical publication is a contract requirement, original art for master assembly drawing illustrations in that publication shall be submitted. Any suitable process, such as silk screening, etc., may be used to produce the required art work, which shall consist of the following:

(a) A board mounted 50 ± 5 percent, screened photo print of the conductor pattern and outlines of separable components on the conductor side of a printed wiring assembly. The view shall be that seen through the printed wiring assembly load from the component side. When the conductor side separable or printed components are to be gray in the (final) printed illustration, in accordance with 3.4.6, they shall be included in this print.

(b) A positive film overlay, on which the component side conductor pattern, printed components and separable components are in 100 percent black dot. The view shall be that seen from the component side of the printed wiring assembly. This film shall be properly sized so that points on the positive film conductor pattern are in exact alignment with through-the-board connecting points on the board mounted photo print, described in (a) above.

(c) Crop marks shall be included on each photo print and associated positive film overlay. Whenever possible, the print and film shall be sized at 100 percent with respect to the printed illustration appearing in the technical publication. In all cases, the photo size reduction required to obtain a reproducible negative, the figure number, title and page number shall be included in this art. When, in ac-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

cordance with 3.2.3, the associated schematic is included in the figure, this information shall be noted on the original art along with sizing and stripping instructions. Such information shall be duplicated on the original art for the schematic diagram.

3.4.8 Master assembly drawing negatives. When the submission of negatives is required by contract or specification, a composite photographic negative of the original art work, described in 3.4.7 of this specification, shall be submitted for each master assembly drawing illustration appearing in a technical publication.

3.5 Maintenance data for printed wiring assemblies. Technical publications covering field and depot maintenance shall include a test procedure and associated voltage measurement table(s) for each type of printed wiring assembly used in an equipment or system. Each test procedure shall specify the type of test equipment required and detailed test set-up instructions, referenced to the associated voltage measurement table(s). The latter shall list the voltages and, when applicable, include illustrations of wave shapes which should be observed at component junctions and test points, under specific test conditions. The information shall be sufficiently detailed to permit the cause of assembly malfunction or failure to be determined through analysis of the test results.

3.6 Maintenance prints. A master assembly drawing, associated schematic diagram, test procedures and voltage measurement tables, described in 3.3 through 3.5 may be combined to form a maintenance print, when such prints are required by contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions applicable to the technical publication in which they appear shall be applicable to printed wiring assembly illustrations, test procedures and tables. In addition, the government reserves the right to perform any inspections deemed necessary to assure compliance with the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking instructions given in specifications applicable to the technical publication in which they appear shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The intent of this specification is to establish detailed requirements for the inclusion of printed wiring assembly illustrations and maintenance data in technical publications. The purpose of establishing these detailed requirements is to insure that technical publications in this category contain the detailed information and illustrations necessary to test and to effect repair of printed wiring assemblies incorporated in certain equipment and systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required (see 5.1).

Preparing activity:
Navy - MC

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:45Z
Check the source to verify that this is the current version before use.

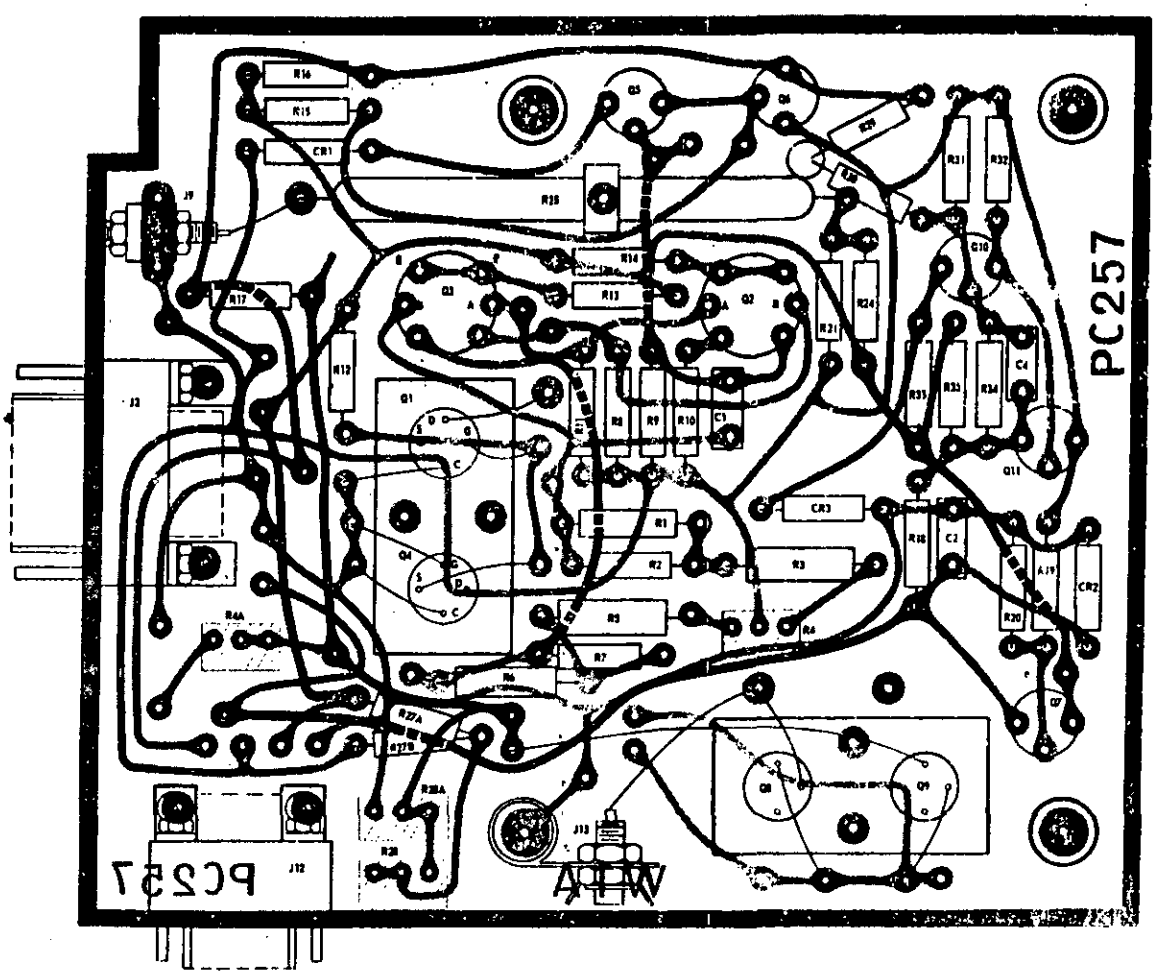


Figure 2. Sample, Master Assembly Drawing

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:45Z
Check the source to verify that this is the current version before use.